Probiotics: Fact Sheet for Consumers (original version)

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Table of Contents

What are probiotics and what do they do?

Probiotics are live microorganisms (such as bacteria and yeasts) that provide health benefits when they are taken in sufficient amounts. They are present in some fermented foods, added to other foods, and available as dietary supplements.

Probiotics act mainly in the digestive tract. Once in the digestive tract, probiotics may improve the function of your digestive system and other aspects of health.

Probiotic microorganisms are named by their genus, species, and strain. An example of a common probiotic microorganism is *Lacticaseibacillus rhamnosus* GG. In this case, *Lacticaseibacillus* is the genus, *rhamnosus* is the species, and GG is the strain. Other common probiotics include specific strains of *Limosilactobacillus reuteri* and *Saccharomyces boulardii*.

What foods provide probiotics?

Microorganisms are used to produce fermented foods, including yogurt, cheese, sourdough bread, pickles, apple cider vinegar, kombucha, miso, kimchi, and sauerkraut. Yogurt, for example, is made by adding live microorganisms (such as *Lactobacillus bulgaricus* and *Streptococcus thermophilus*) to milk. However, while microorganisms are used to make these foods, not all these foods necessarily contain probiotics with proven health benefits.

Manufacturers will sometimes add probiotics to unfermented foods, such as cereals, juices, milks, nutrition bars, smoothies, and infant and toddler formulas. If a food contains probiotics, the genus, species, and strain of the probiotic should be listed on the product label.

What kinds of probiotic dietary supplements are available?

Dietary supplements that are labeled as probiotics may contain different microorganisms, and the number of microorganisms they contain can vary widely between products. Because some of these supplements may not have been studied, any potential health effects they have may not be known.

The Supplement Facts label on a dietary supplement that contains probiotics lists the total weight of the microorganisms in the product. Many product labels also list the number of colony-forming units (CFUs) in a serving. The number of CFUs is a more accurate representation of how many active microorganisms are in a product than the total weight of the microorganisms. Examples of amounts of CFUs that you might see on a label are 1 x 10^9 (1 billion) CFUs and 1 x 10^{10} (10 billion) CFUs. Higher CFU counts alone do not necessarily mean that the product has greater health benefits. A product's health benefits, if any, are determined by both the specific microorganisms (or strains) it contains and the number of CFUs.

What are some possible effects of probiotics on health?

Scientists are studying probiotics to understand how they affect health. Here are some examples of what this research has shown.

Atopic dermatitis

Atopic dermatitis (eczema) is a skin condition that mostly affects children. People with atopic dermatitis have patches of dry, itchy skin. They also experience red rashes that can come and go, and these rashes may develop sores that ooze or bleed when scratched. Some studies have shown that when pregnant women take probiotics and their infants are given probiotics after birth, the infants might have a lower risk of developing atopic dermatitis and experience less severe symptoms of dermatitis. However, the effects vary depending on the probiotic strain used and whether the mother takes it during pregnancy, the infant is given it after birth, or both.

Pediatric acute infectious diarrhea

Acute infectious diarrhea in infants and children causes loose or liquid stools and three or more bowel movements within 24 hours. This condition is often caused by a viral infection and can last for up to a week. Some infants and children also develop fever and vomiting. Some studies have shown that certain probiotics shorten bouts of acute diarrhea by about 1 day. Other studies, however, have not shown that probiotics are effective.

Antibiotic-associated diarrhea

Antibiotics, such as erythromycin and penicillin, can kill beneficial microorganisms that live in the digestive tract, which sometimes results in diarrhea. Some probiotic strains might help reduce the risk of antibiotic-associated diarrhea in children and adults. This is especially true when people start taking these products within 2 days of the first antibiotic dose.

Necrotizing enterocolitis

Necrotizing enterocolitis (NEC) is a serious and sometimes deadly digestive tract illness that mostly affects preterm babies, especially those who are very small. Some probiotic strains might help reduce the risk of NEC in preterm infants. While these products are sometimes used in hospitals, additional research on their safety and effectiveness is needed because the U.S. Food and Drug Administration (FDA) has raised some concerns about their use in preterm infants.

Inflammatory bowel disease

Inflammatory bowel disease (IBD) is a chronic disease that can affect children and adults. Types of IBD include ulcerative colitis and Crohn's disease. People with IBD commonly have diarrhea, stomach pain, or bloody stools that are caused by chronic inflammation in the digestive tract. Taking probiotics with medications might slightly reduce the severity of the symptoms of ulcerative colitis, but it does not seem to help people with Crohn's disease.

Irritable bowel syndrome

Irritable bowel syndrome (IBS) is a common disorder that can affect children and adults, especially women. It causes frequent stomach pain and discomfort, bloating, changes in bowel movement frequency, and diarrhea or constipation. The causes of IBS are unclear, but this condition may be a result of changes in the types of microorganisms in the digestive tract. Taking probiotics might help manage IBS symptoms.

However, the effects vary depending on the probiotic strain used, how long it is used, and the symptom being treated.

Hypercholesterolemia

Very high levels of cholesterol in the blood (a condition known as hypercholesterolemia) and a buildup of cholesterol in blood vessel walls can block the flow of blood to the heart and increase the risk of cardiovascular disease. This condition can affect people of all ages, but it is most common in people over 40 years of age. Some studies have shown that probiotics can slightly lower the levels of total cholesterol and low-density lipoprotein (LDL) cholesterol, but other studies have found no benefits. More research is needed to understand the effects of probiotics on blood cholesterol.

Obesity

Some studies have shown that probiotics might slightly reduce body weight or body fat. Other studies have shown that probiotics have no effect or might even increase body weight. More research is needed to understand the effects of probiotics on body weight and body fat.

Can probiotics be harmful?

Most probiotics have been safely used in foods or dietary supplements for decades. Some of the same types of probiotics have been present in fermented food for thousands of years. In healthy people, probiotics may cause gas, but they rarely cause infections or other health problems.

However, probiotics might not be safe for everyone. The FDA has reported that probiotics might cause infections or even life-threatening illness in preterm infants. Probiotics might also cause problems such as bacterial infections in people who are already seriously ill or who have weak immune systems.

What should I know about choosing a probiotic?

There are no official recommendations that cover the use of probiotics by healthy people. If you want to try probiotics, ask your health care provider for advice about which probiotic to choose, what dose to take, and how long to use the product. On the product label, you can find information about the genus, species, and strain of the microorganisms; the expiration or use-by date; and storage instructions. Some probiotics need to be kept in the refrigerator, but others can be stored at room temperature.

The product label will also indicate the number of CFUs in the product. Some products list the number of CFUs at the time of manufacture whereas others list the number through the expiration date or use-by date. It is worth noting that the number of CFUs in a probiotic can decline over time. Therefore, the number of CFUs that a product contained when it was manufactured does not necessarily reflect the number of CFUs that the product contains when it is purchased or used. For this reason, some experts recommend choosing products that list the number of CFUs that the product will contain by the expiration date or use-by date instead of the number it contained at the time of manufacture.

Where can I find out more about probiotics?

- For more information on probiotics
    - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on Probiotics
    - National Center for Complementary and Integrative Health, Probiotics
    - United States Department of Agriculture (USDA) Agricultural Research Service, Keeping a Healthy Gut
- For more advice on buying dietary supplements
    - ODS Frequently Asked Questions: Which brand(s) of dietary supplements should I purchase?
- For information on the government's food guidance system
    - Dietary Guidelines for Americans

- [MyPlate]

Glossary

acute
: Sudden, severe, and not long lasting.

association
: A relationship between two conditions or states such that if one is present, the other is likely to be present as well. An association between two conditions or states, however, does not necessarily imply a cause and effect relationship. The terms association and relationship are often used interchangeably.

bacteria
: Single-celled organisms that are too small to be seen without a microscope. Bacteria are found everywhere and may be helpful or harmful.

blood vessel
: A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

cardiovascular disease
: CVD. A general term referring to disorders of the heart and blood vessels. CVD includes coronary artery disease, heart failure, atherosclerosis, high blood pressure, peripheral artery disease, and stroke.

cholesterol
: A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

chronic
: Happening for a long time, persistently, or repeatedly.

chronic disease
: A condition that is continuous or recurrent, is not easily cured, and cannot be passed from person to person. Examples of chronic diseases include heart disease, diabetes, and asthma.

constipation
: A condition in which stool becomes hard, dry, and difficult to pass and bowel movements happen infrequently. Other symptoms may include painful bowel movements and feeling bloated, uncomfortable, and sluggish.

diarrhea
: Loose, watery stools.

dietary supplement
: A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

digestive tract
: The large, muscular tube that extends from the mouth to the anus, in which hormones, enzymes, and the movement of muscles work together to digest food. Also called the gastrointestinal (GI) tract.

disorder
: In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose
: The amount of medicine or other substance taken at one time or over a specific period of time.

genus
: The name of a category that is part of the scientific classification of all organisms. Genus is located in the classification system after kingdom, phylum, class, order, and family and before the subclassification of species. Humans, for example, belong to the genus *Homo* and the species *Homo sapiens*.

health care provider
: A person who supplies health care services. Health care providers include individuals with professional training (including doctors, nurses, technicians, and aides).

immune system

A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant
: A child younger than 12 months old.

infection
: The invasion and spread of germs in the body. The germs may be bacteria, viruses, yeast, or fungi.

inflammation
: Redness, swelling, pain, and/or a feeling of heat in an area of the body. It is a protective reaction to injury, disease, or irritation of tissues.

inflammatory bowel disease
: IBD. Long-lasting (chronic) problems that cause irritation and ulcers in the digestive tract. The most common disorders are ulcerative colitis and Crohn's disease.

label
: When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

microorganism
: A living being that can be seen only through a microscope. Microorganisms include helpful and harmful bacteria, protozoa, algae, and fungi. Although viruses are not considered living organisms, they are sometimes classified as microorganisms.

nutrition
: The process of eating, digesting, and absorbing nutrients (such as protein, carbohydrate, fat, vitamins, minerals, and water) from food to maintain the body, grow new cells, repair tissues, and supply energy. Nutrition is also the science of food, diet, and health.

Office of Dietary Supplements
: ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

risk
: The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

species
: The name of a category that is part of the scientific classification of all organisms. The category species is located in the classification system after kingdom, phylum, class, order, family and genus. Humans, for example, belong to the genus *Homo* and the species *Homo sapiens*.

stool
: The waste matter passed in a bowel movement; feces.

supplement
: A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom
: A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

toddler
: A child between the ages of 12 months and 3 years.

ulcerative colitis
: Chronic inflammation of the colon that causes ulcers to form in its lining. This condition is marked by abdominal pain, cramps, and loose discharges of pus, blood, and mucus from the bowel.

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